



American International University-Bangladesh
Faculty of Engineering (FE)
Department of Electrical and Electronic Engineering (EEE)

ELECTRONIC DEVICES LAB

SECTION: Y

Course Teacher:

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Submitted By:

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Lab Report:3

1. Title: *Study of Diode Clipping and Clamping Circuits.*

2. Abstract:

The main objective of this experiment was to know about the clipper circuits and clamper circuits. Nine circuits corresponding to Figure 1 to 9 were analyzed and total nine table of experimental data were recorded. Input and output waveforms of sinusoidal, triangular and rectangular shapes were observed using oscilloscope. For this experiment, Ni Multisim was used to draw the simulation of circuits.

3. Theory:

Clipper

A clipper is an electronic circuit designed to prevent the output of a circuit from exceeding a predetermined voltage level. Clippers clip away a portion of an input signal. There are two general categories of clippers:

1. Series clipper.
2. Parallel clipper.

Series Clipper: The series configuration is defined as one where the diode is in series with the load as shown in Fig. 1 (a). This clipper is again divided into two types:

- **Unbiased series clipper:** Figure 1 (a) and (b) shows the unbiased series clipper circuit and the response of the unbiased series configuration, where no DC supply is connected in series with the diode.
- **Biased series clipper:** Figure 2 (a) and (b) shows the biased series clipper circuit and the response of the biased series configuration where a DC supply is connected in series with the diode.

Parallel Clipper: The network of Fig. 3 (a) is the simplest of parallel diode configurations. The parallel clipper has the diode in a branch parallel to the load. This clipper is again divided into two types:

- **Unbiased parallel clipper:** Figure 3 (a) shows the unbiased parallel clipper circuit and the response of the unbiased parallel configuration of Fig. 3 (a) is provided in Fig. 3 (b).
- **Biased parallel clipper:** Figure 4 (a) (b) shows the biased parallel clipper circuit and response of the circuit. Figure 5 (a) shows the biased parallel positive clipper circuit, where a DC supply is connected in series with the diode to clip off the positive half-cycle. Figure 5 (b) shows the biased parallel negative clipper circuit, which clips off the negative half-cycle. Figure 5 (c) shows the biased parallel dual clipper circuit, where a DC supply is connected in series with the diode to clip off both the positive and negative half-cycles.

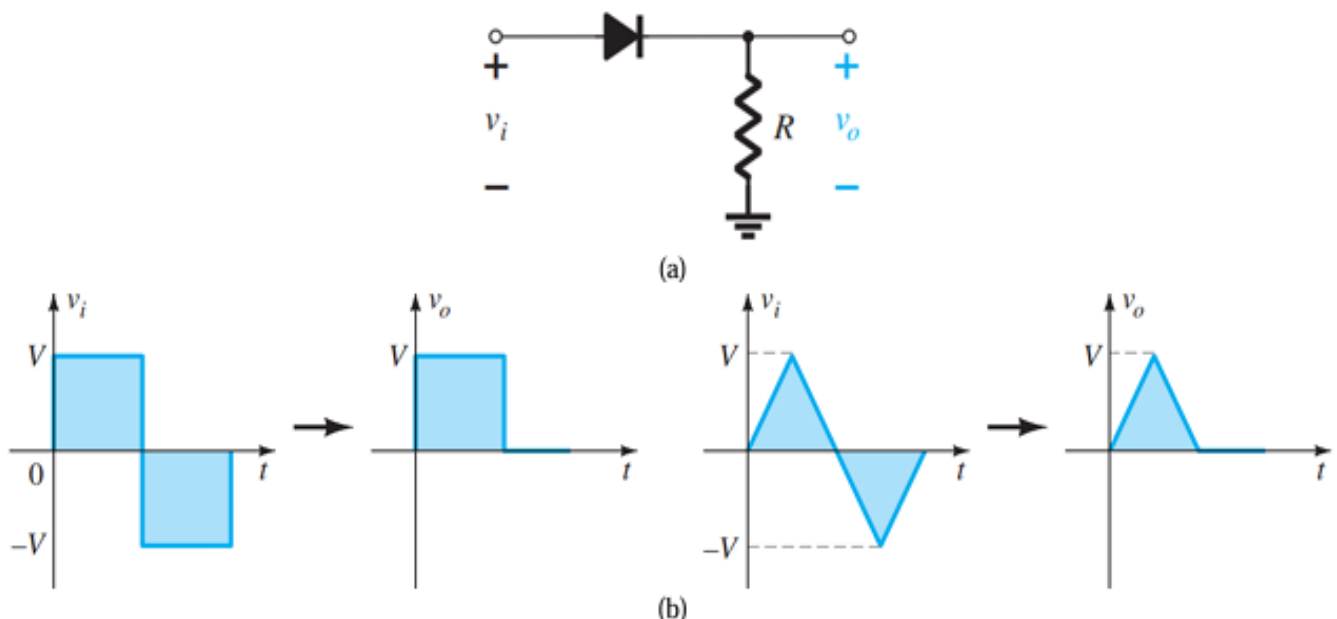


Figure 1: Unbiased series clipper: (a) Circuit diagram; (b) Input-output wave shapes (rectangular and triangular).

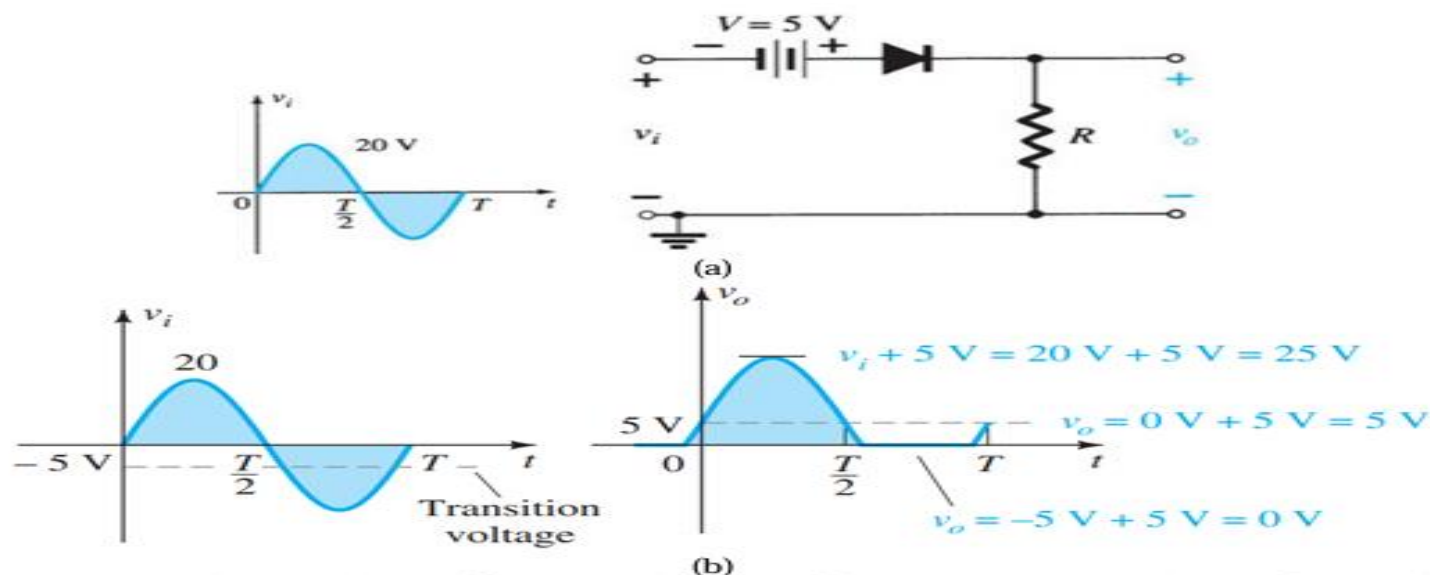


Figure 2: Biased series clipper: (a) Circuit diagram; (b) Input-output wave shapes (sinusoidal).

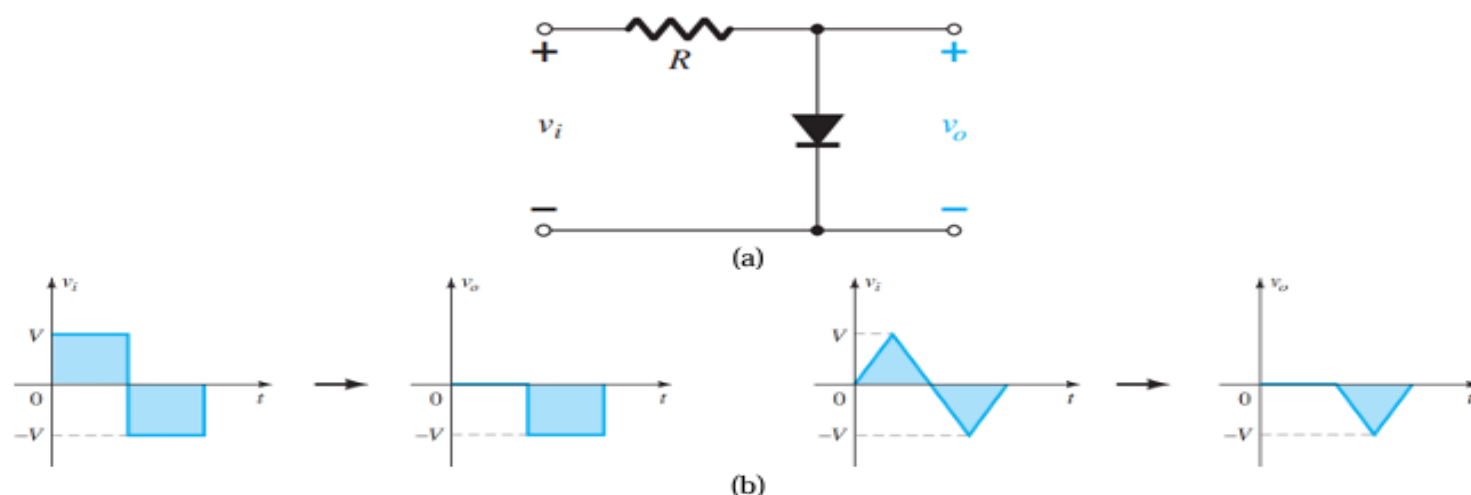


Figure 3: Unbiased parallel clipper: (a) Circuit diagram; (b) Input-output wave shapes (rectangular and triangular).

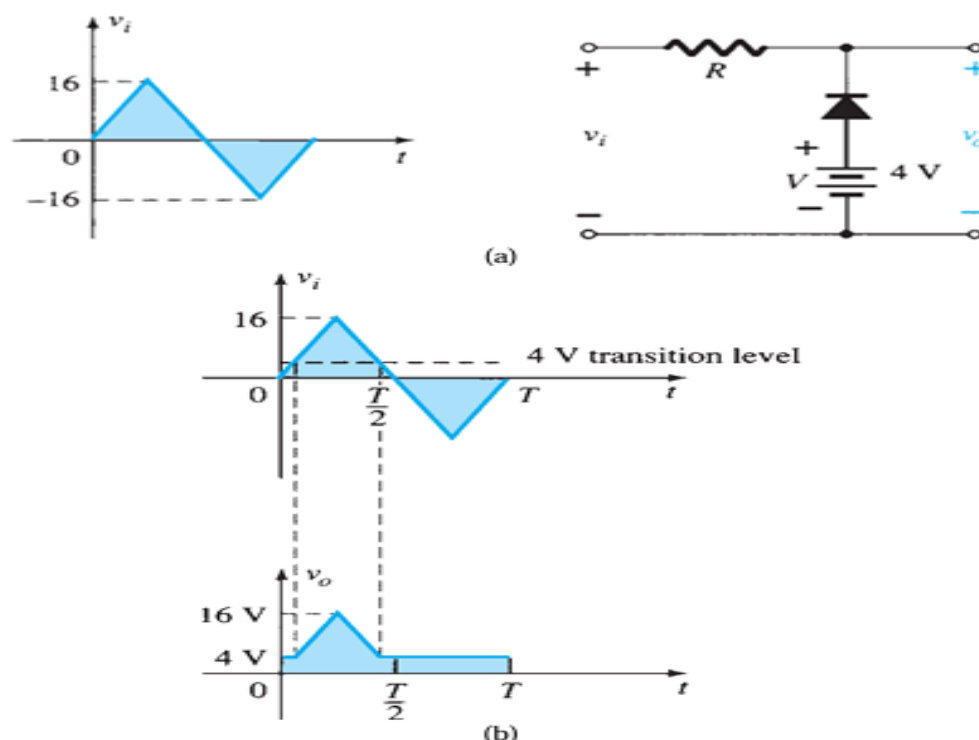


Figure 4: Biased parallel clipper: (a) Circuit diagram; (b) Input-output wave shapes (triangular).

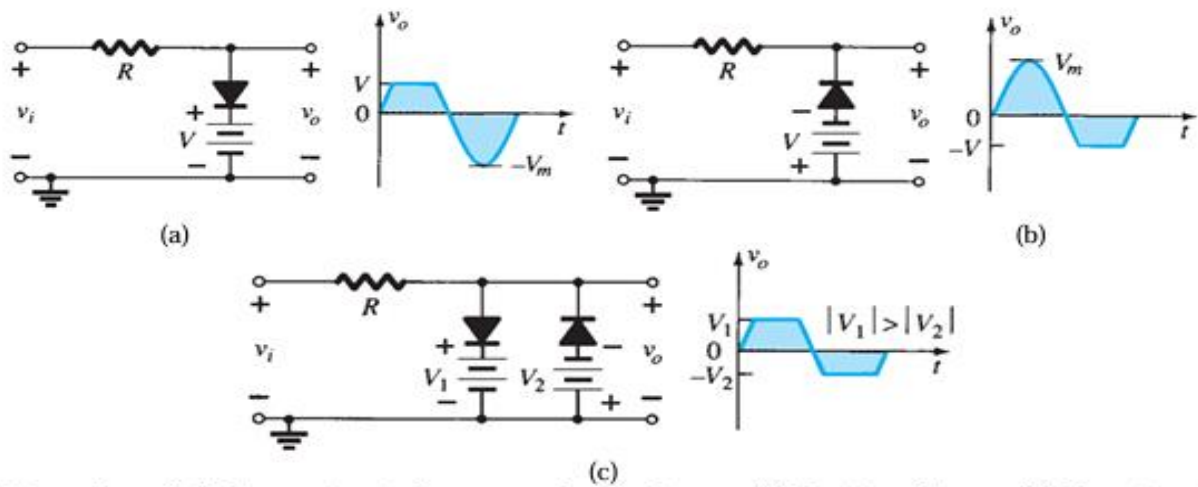


Figure 5: Biased parallel clipper circuit diagram and waveshapes: (a) Positive clipper; (b) Negative clipper; (c) Both sides clipper.

Clamper:

A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different DC level without changing the appearance of the applied signal. There are two types of clamper circuits:

1. Positive clamper circuit
2. Negative clamper circuit.

Each type again may be divided into further two types:

- **Unbiased positive clamper:** An unbiased positive clamper circuit adds positive DC voltage as shown in Fig. 6.
- **Biased positive clamper:** It takes the DC level to a more positive direction depending on the applied DC voltage connected in series with the diode as shown in Fig. 7 (a) and (b).
- **Unbiased negative clamper:** An unbiased negative clamper circuit adds negative DC voltage as shown in Fig. 8.
- **Biased negative clamper:** The biased negative shown in Fig. 9 (a).

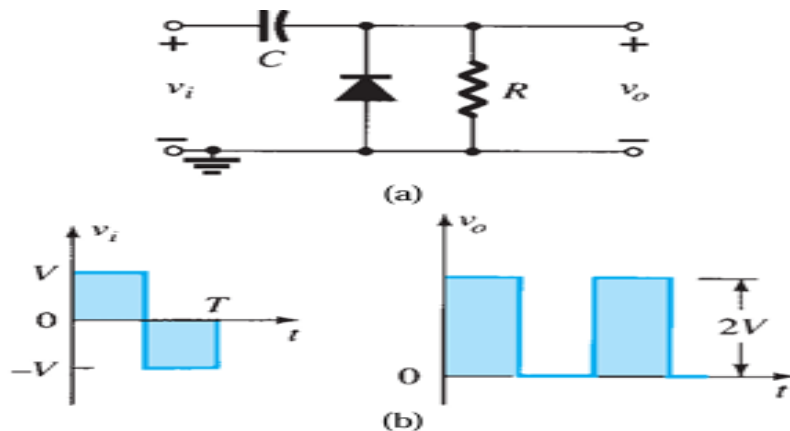


Figure 6: Unbiased positive clamper: (a) Circuit diagram; (b) Input-output wave shapes (square).

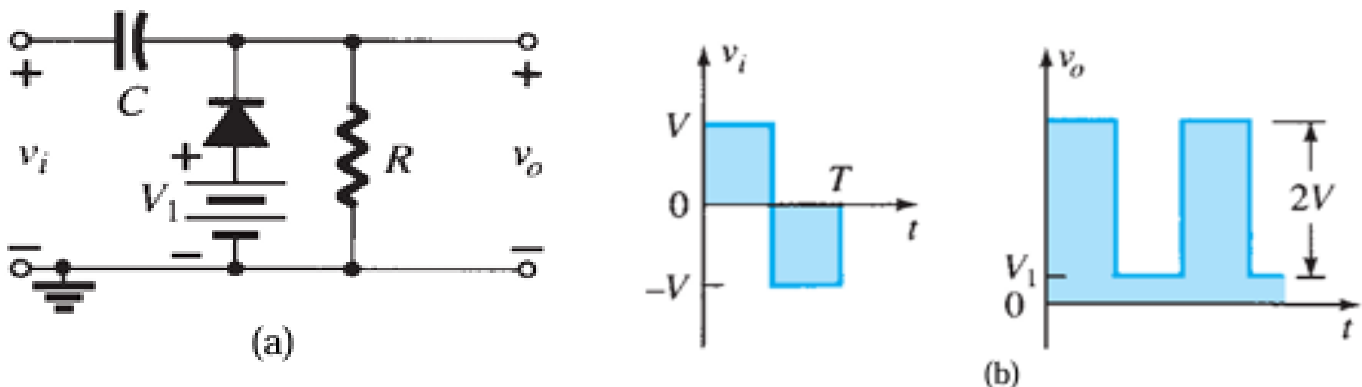


Figure 7: Biased positive clamper: (a) Circuit diagram; (b) Input-output wave shapes (square).

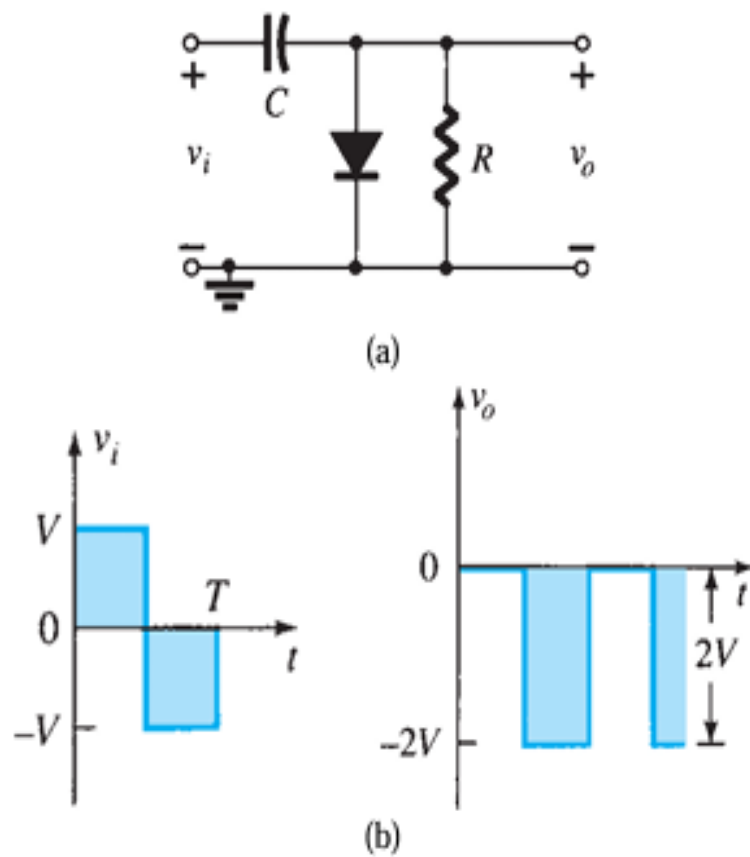


Figure 8: Unbiased negative clipper: (a) Circuit diagram; (b) Input-output wave shapes (square).

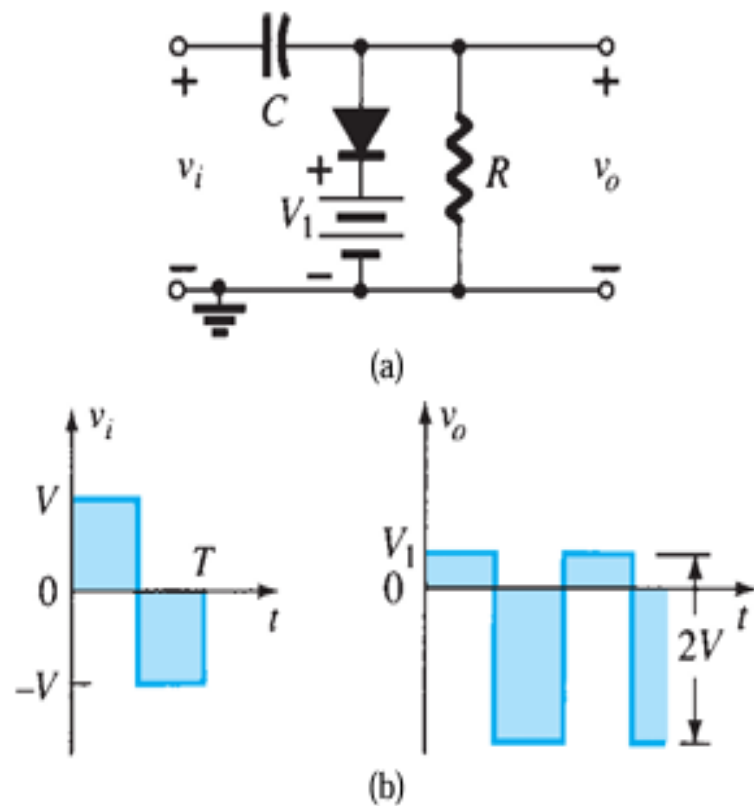
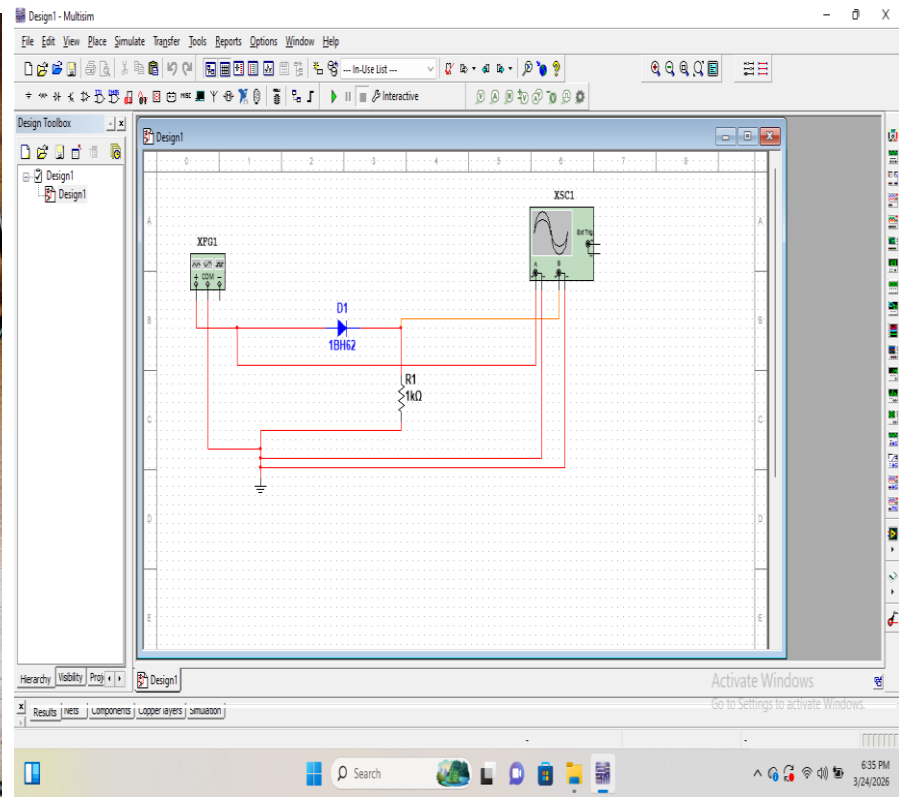


Figure 9: Biased negative clipper: (a) Circuit diagram; (b) Input-output wave shapes (square).

4. Methodology:



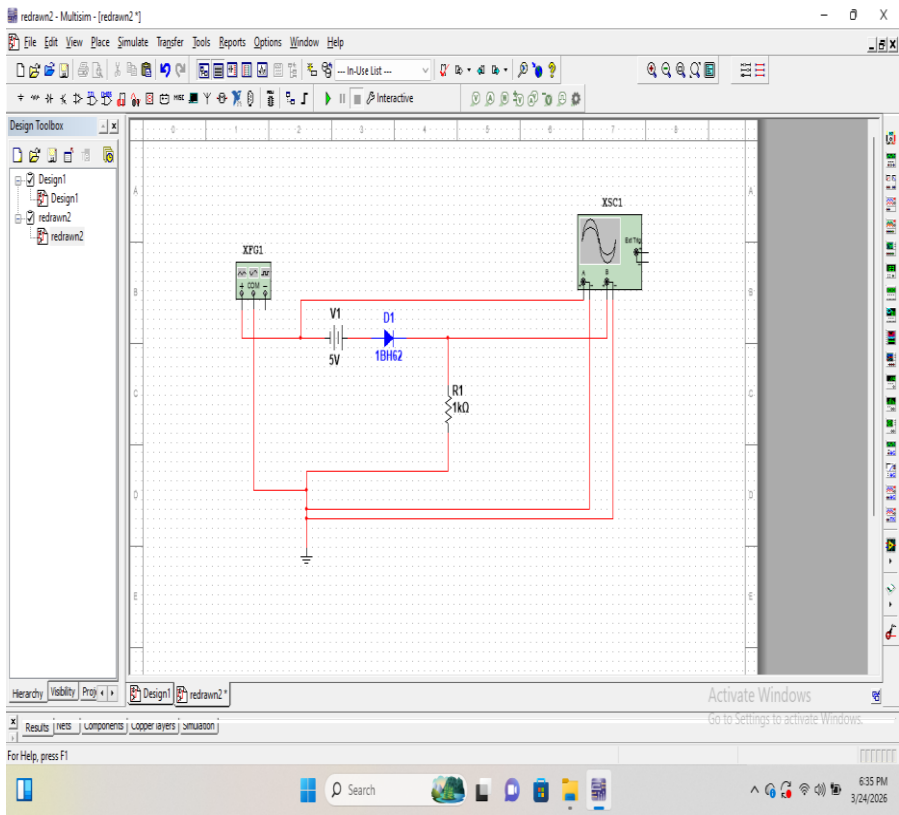
Hardware diagram- Table:1



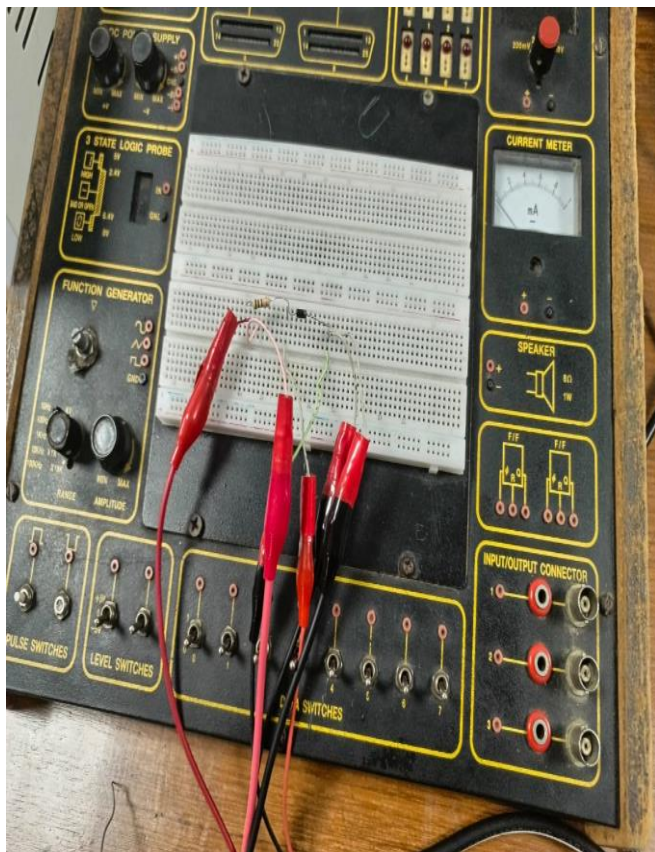
Simulation diagram- Table:1



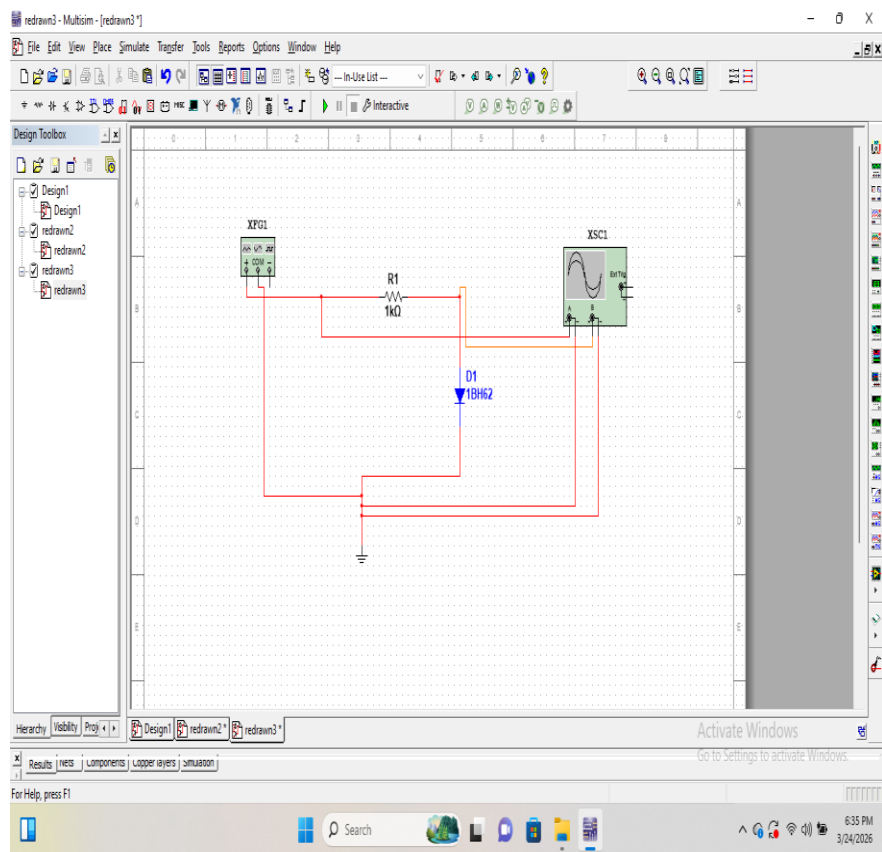
Hardware diagram- Table:2



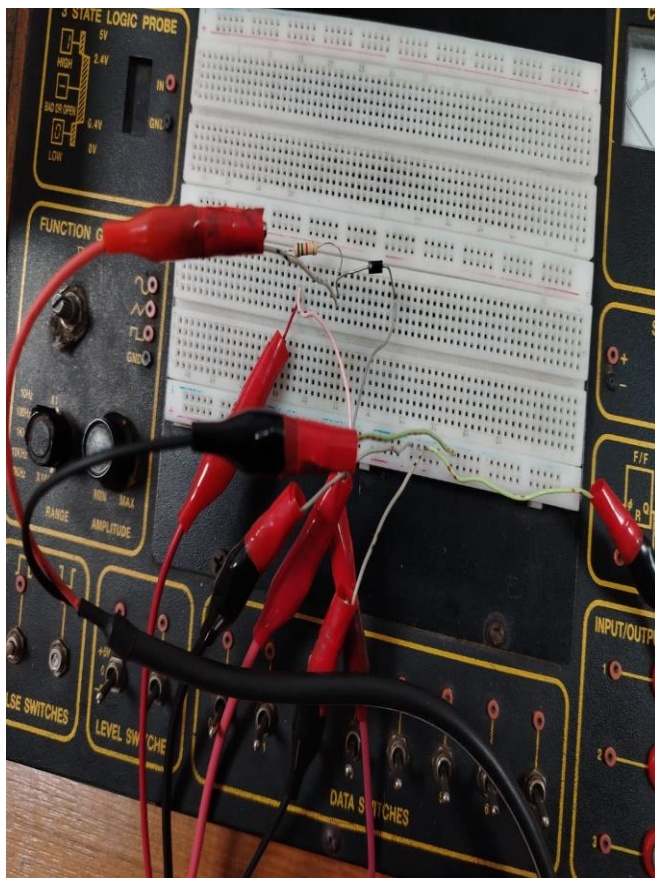
Simulation diagram- Table:2



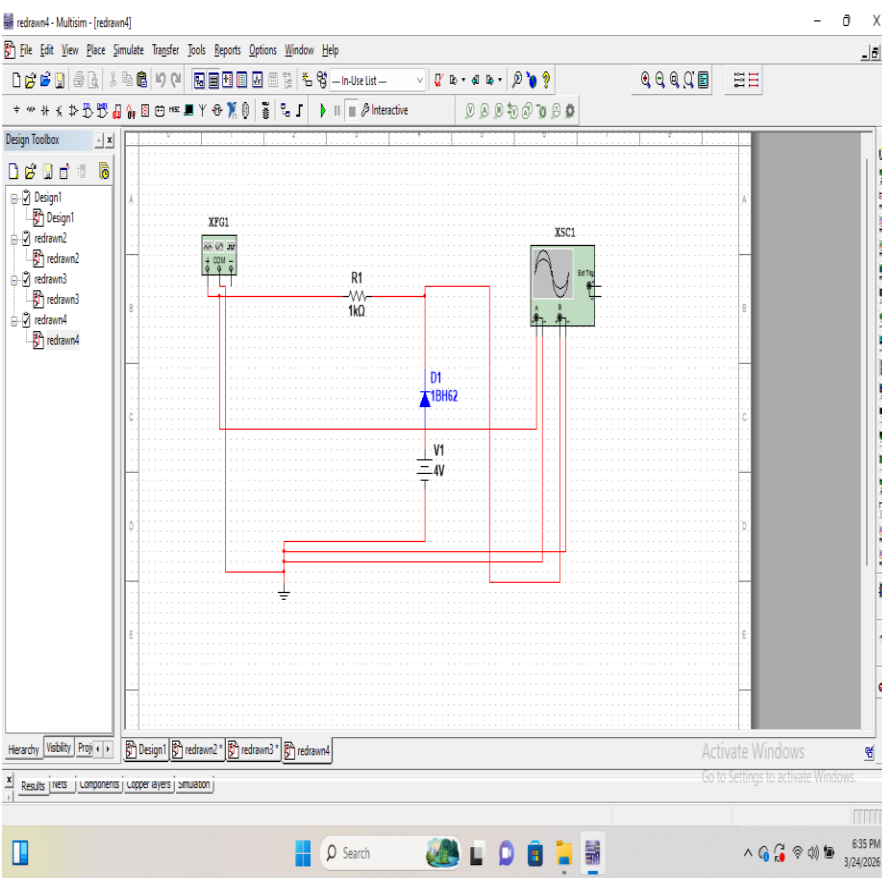
Hardware diagram- Table:3



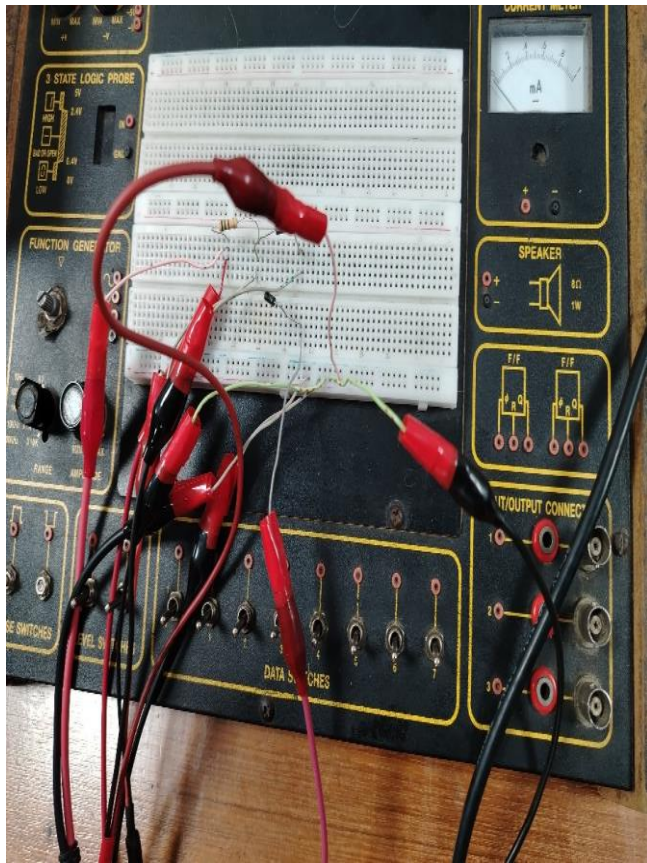
Simulation diagram- Table:3



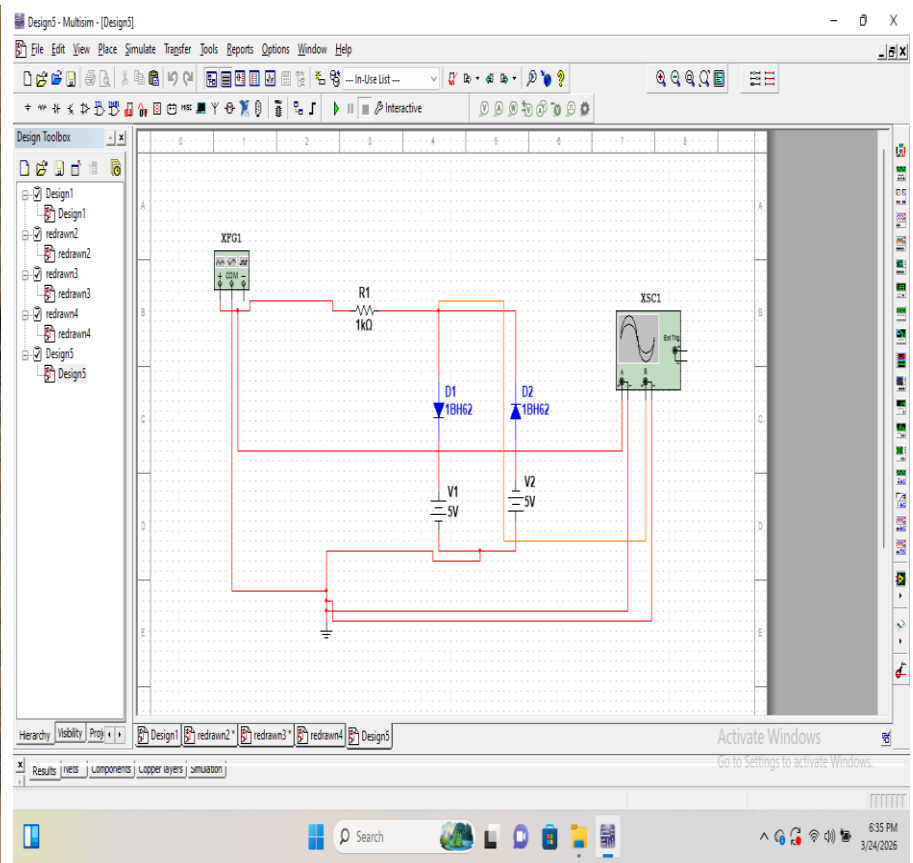
Hardware diagram- Table:4



Simulation diagram- Table:4



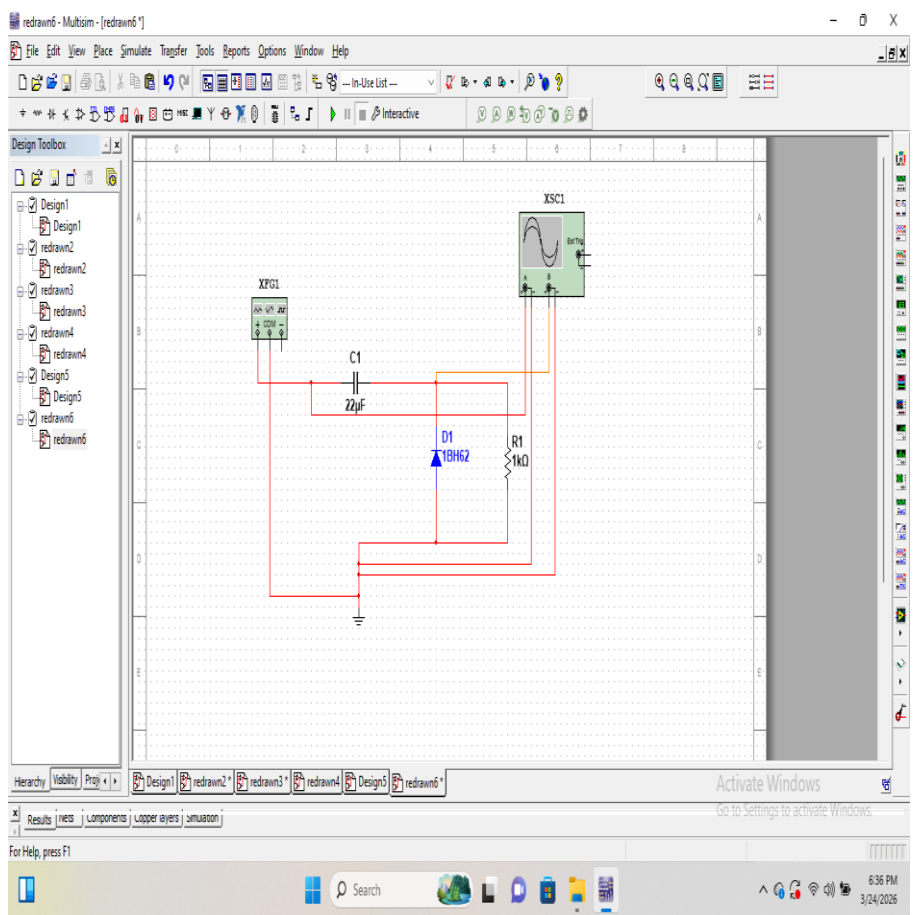
Hardware diagram- Table:5



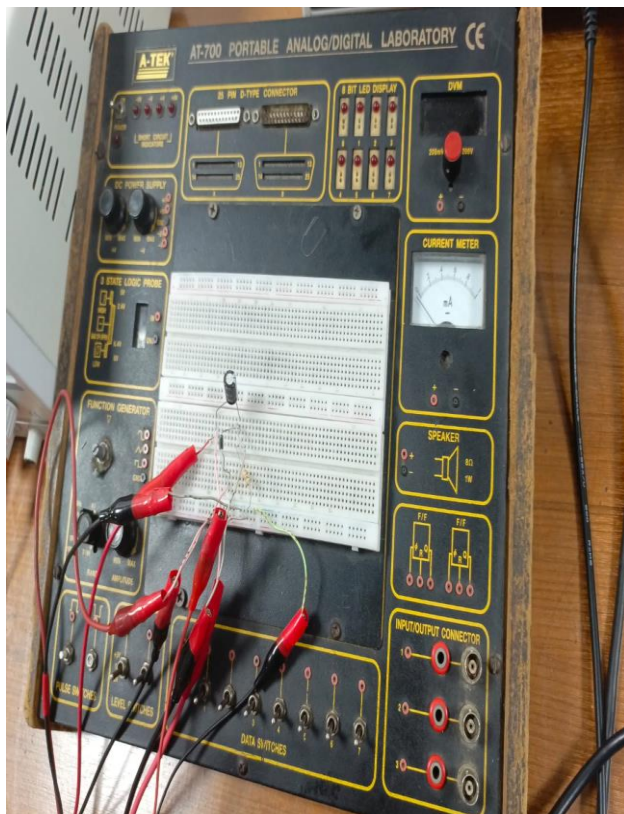
Simulation diagram- Table:5



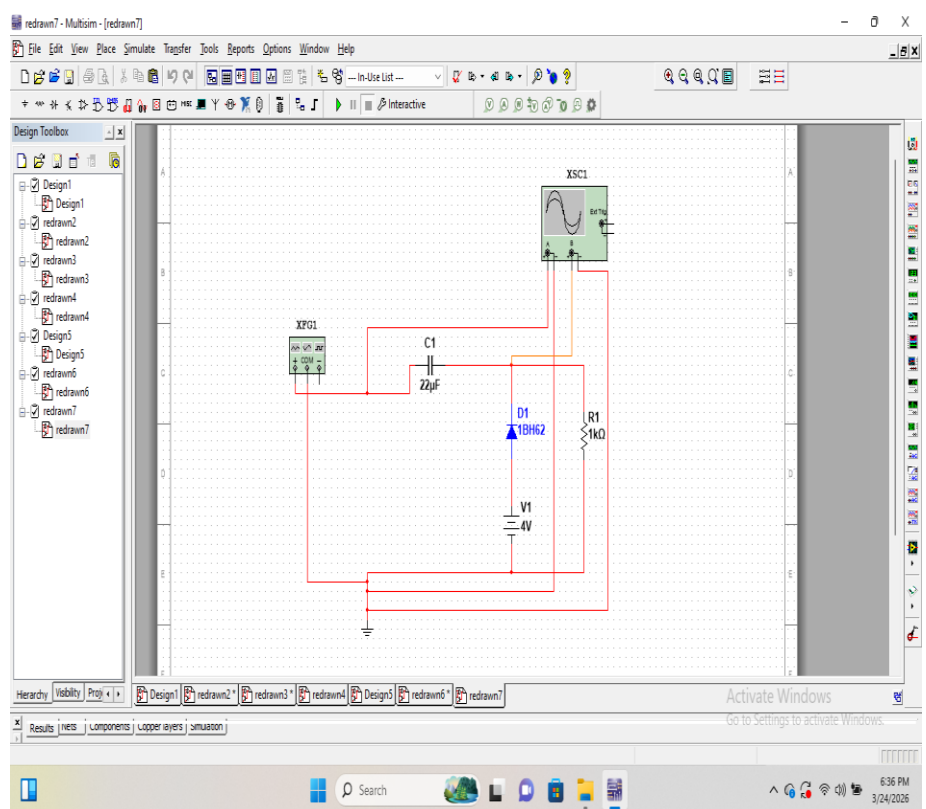
Hardware diagram- Table:6



Simulation diagram- Table:6



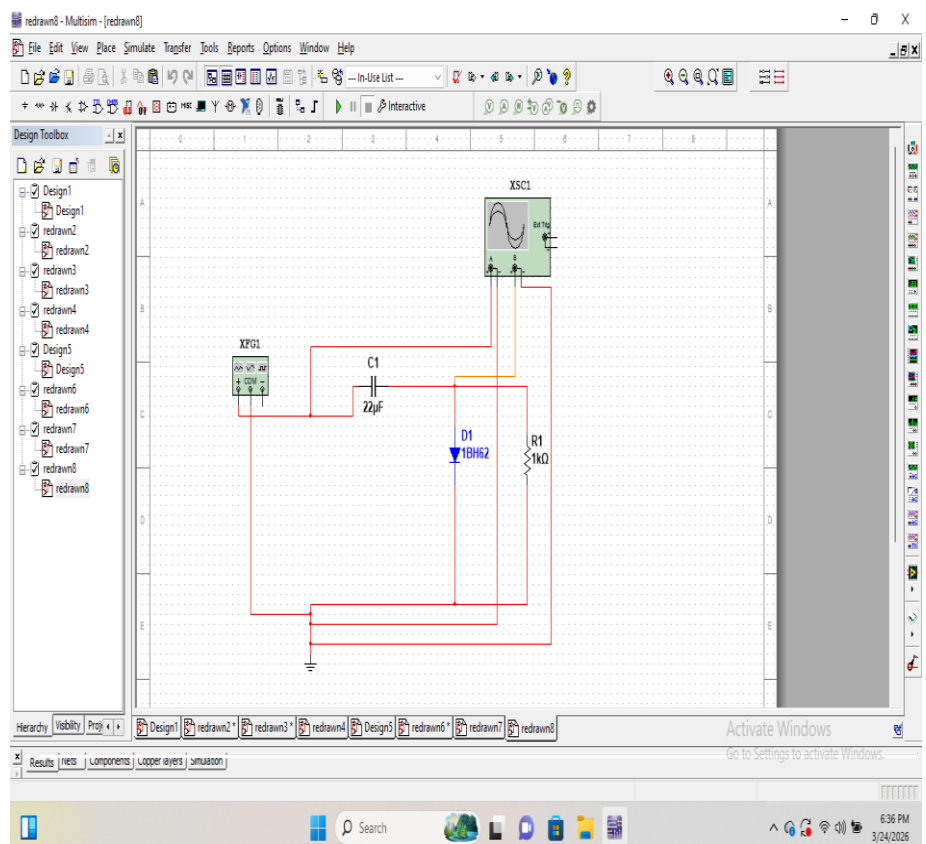
Hardware diagram- Table:7



Simulation diagram- Table:7



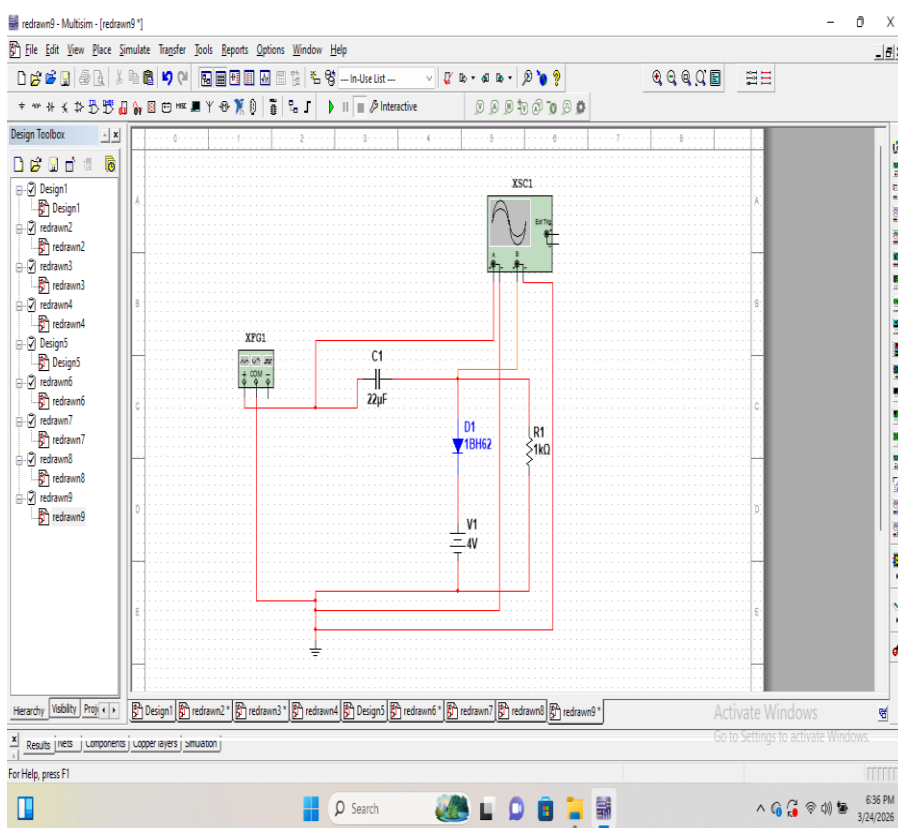
Hardware diagram- Table:8



Simulation diagram- Table:8



Hardware diagram- Table:9

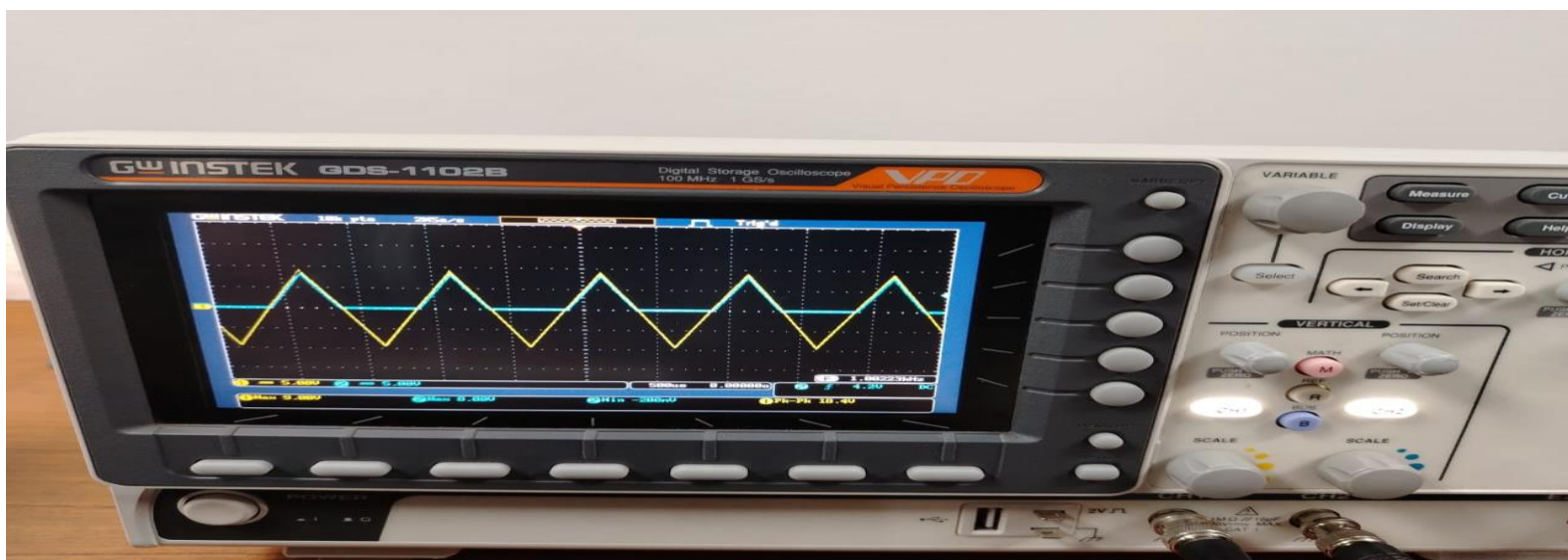


Simulation diagram- Table:9

5. Results and Analysis:

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Sinusoidal	9.60	8.80	9.68	8.92
Triangular	9.00	8.00	10.00	9.23
Rectangular	9.80	8.80	10.00	9.23

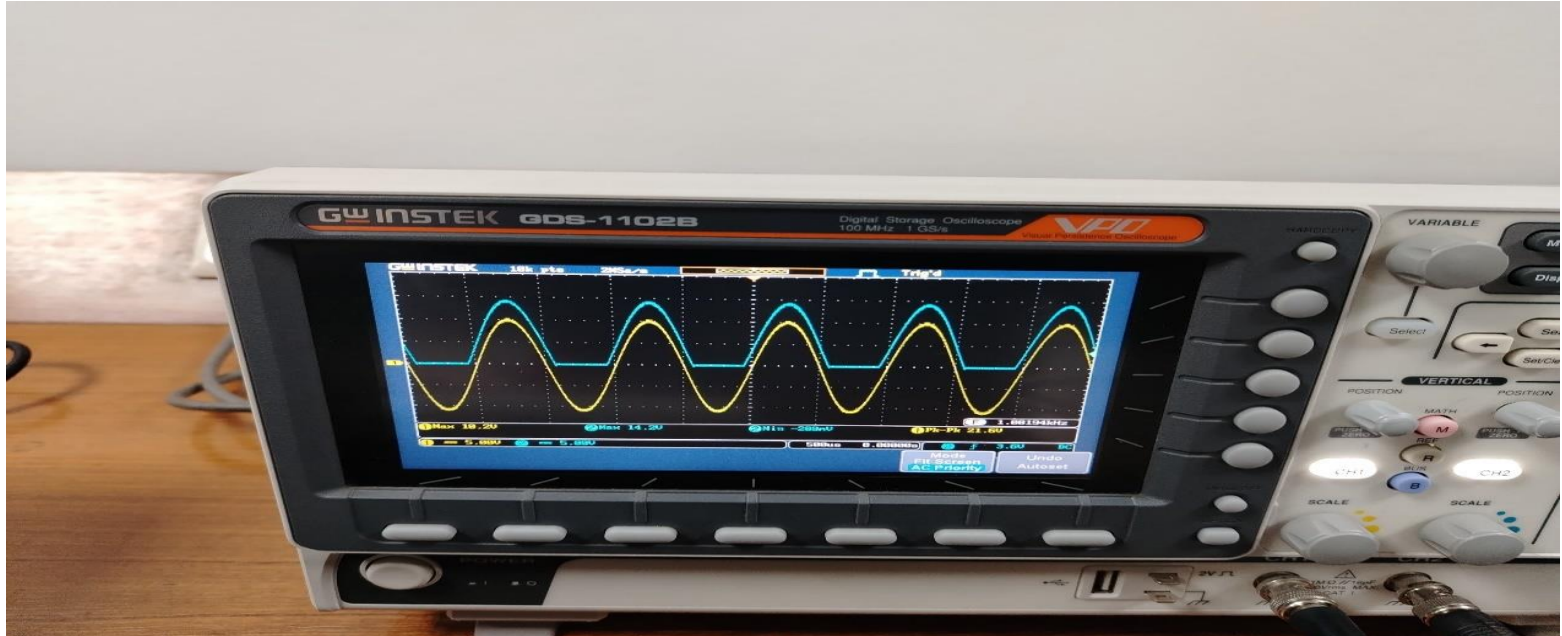
Table 1: Data Table for the Circuit of Figure 1 (a)



Oscilloscope output- Table:1

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Sinusoidal	10.2	14.2	9.76	13.95
Triangular	9.4	13.6	9.60	13.79
Rectangular	10.4	14.2	10.00	14.21

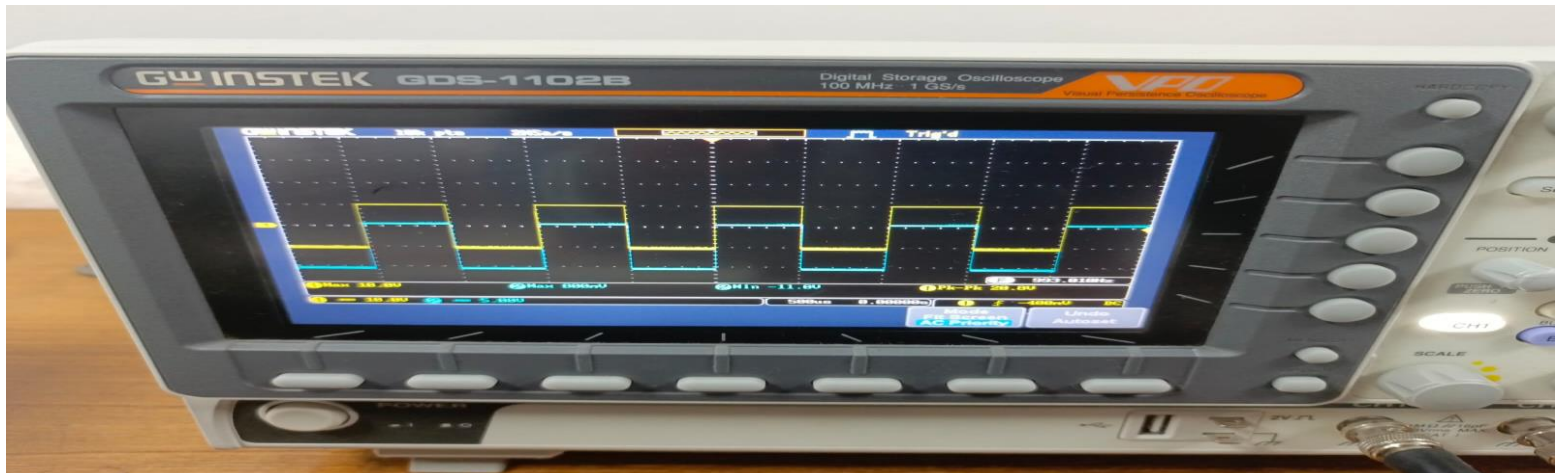
Table 2: Data Table for the Circuit of Figure 2 (a)



Oscilloscope output- Table:2

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (mV) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Sinusoidal	10.0	800	9.91	765.48
Triangular	9.60	600	9.60	761.99
Rectangular	10.8	800	10.00	747.46

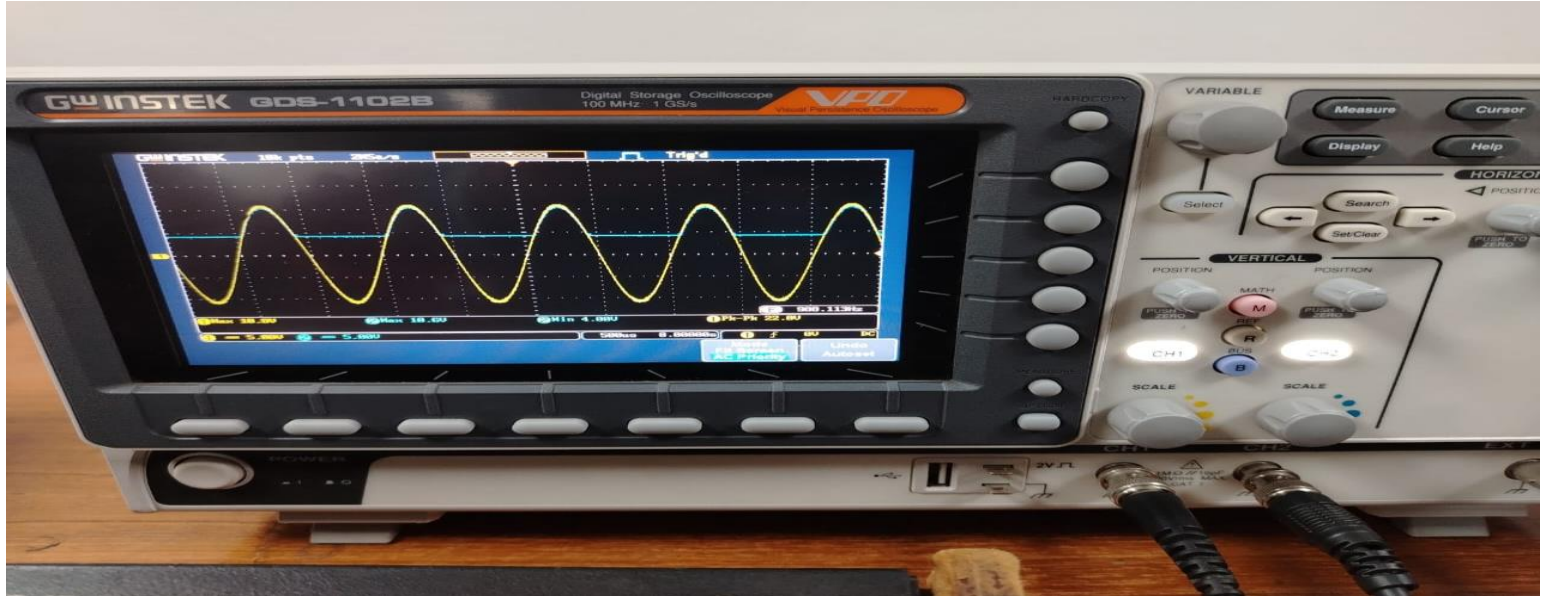
Table 3: Data Table for the Circuit of Figure 3 (a)



Oscilloscope output- Table:3

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Sinusoidal	10.8	10.6	9.30	9.30
Triangular	10.4	10.2	10.00	10.01
Rectangular	11.2	11.0	10.00	9.99

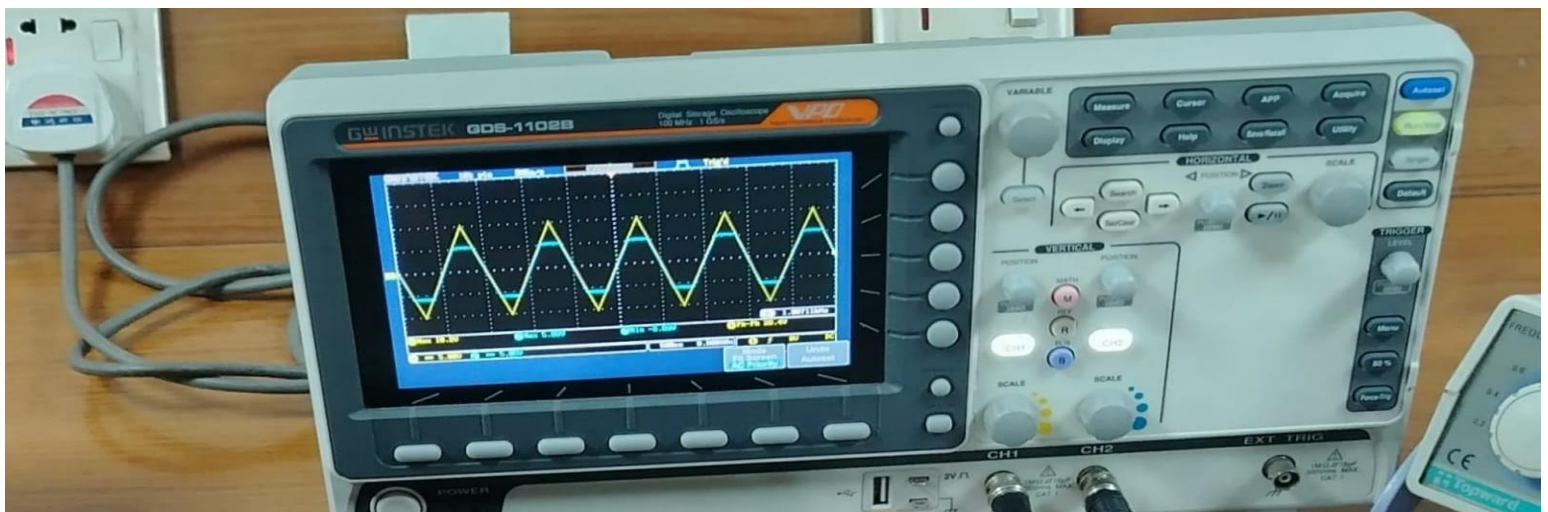
Table 4: Data Table for the Circuit of Figure 4 (a)



Oscilloscope output- Table:4

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Sinusoidal	10.8	5.60	9.94	5.69
Triangular	10.2	5.60	10.00	5.69
Rectangular	10.8	5.80	10.00	5.69

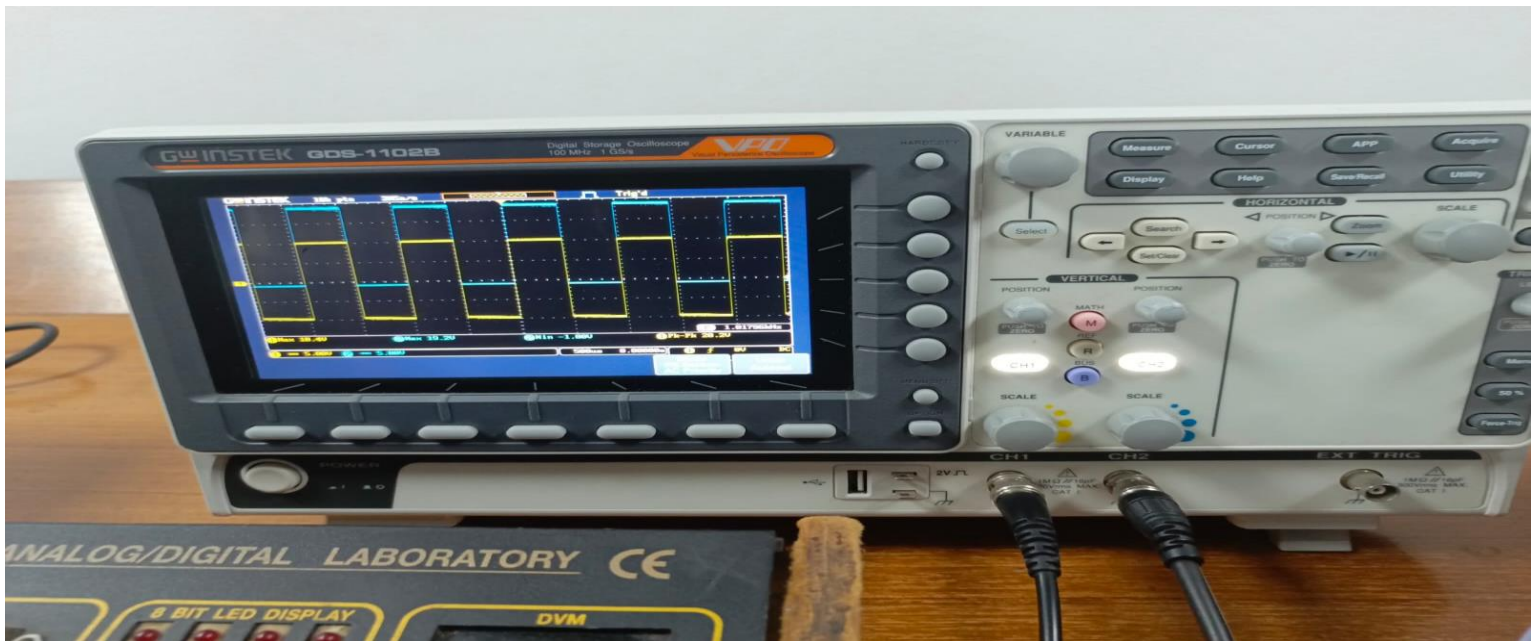
Table 5: Data Table for the Circuit of Figure 5 (c)



Oscilloscope output- Table:5

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Rectangular	10.4	19.2	10.00	19.32

Table 6: Data Table for the Circuit of Figure 6 (a)



Oscilloscope output- Table:6

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Rectangular	10.0	22.0	10.00	22.75

Table 7: Data Table for the Circuit of Figure 7 (a)



Oscilloscope output- Table:7

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Rectangular	10.0	5.20	10.00	1.11

Table 8: Data Table for the Circuit of Figure 8 (a)



Oscilloscope output- Table:8

Wave Shape	Hardware values		Simulated values	
	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (mV) (From Oscilloscope)	Peak Input Voltage, V_i (V) (From Oscilloscope)	Peak Output Voltage, V_o (V) (From Oscilloscope)
Rectangular	10.0	800	10.00	4.97

Table 9: Data Table for the Circuit of Figure 9 (a)



Oscilloscope output- Table:9

6. Discussion:

The main objective of this experiment was to study how the diode, along with resistors, capacitors and biasing sources, modifies the input waveform by clipping or shifting it.

Both the hardware implementations and simulation of circuits were shown. For simulation purpose, Ni Multisim was used. Different waveforms such as sinusoidal, triangular and rectangular signals were applied as input and the corresponding output voltages were observed and recorded.

In the clipper circuits (Figures 1-5), it was observed that specific portions of the input waveform were removed depending on the orientation of the diode and the applied bias voltage. The output voltages in both hardware and simulation were close to each other, with minor differences. For example, in case of unbiased series clipper (Figure 1), the output voltages were slightly lower than the input due to the forward voltage drop across the diode. In case of Figure 2 and 5, the effect of biasing was clearly visible, where the clipping level shifted according to the applied DC voltage, and the results closely matched theoretical expectations.

In Figure 3, the output voltages significantly reduced to the millivolt range, which indicated strong clipping action. Both hardware and simulation result showed similar behavior, confirming correct circuit operation. In Figure 4, the output voltage remained close to the input, indicating minimal clipping, which also matched expected behavior. In the clamper circuits (Figures 6-9), the waveform shifted vertically without altering shape. The output voltages were higher than the input in some cases, showing the effect of DC level shifting. The simulation results were slightly higher than the hardware values, which is expected due to ideal component behavior in simulation. In Figure 8 and Figure 9, the outputs showed lower or reduced voltage levels, indicating proper clamping and diode action.

Some differences between hardware and simulated results were observed. These differences occurred due to several factors. In simulation, ideal components were used, which do not include real-world effects. In simulation, ideal components were used, which do not include real-world effects such as internal resistance, exact forward voltage variation and leakage current. In contrast, real hardware components exhibit non-ideal behavior, causing voltage drops and slight deviations. Additionally, measurement errors from oscilloscope readings, noise and environmental conditions also contributed to variation in hardware results. In some cases, differences in units (mV vs V) and scaling also created apparent mismatches.

Furthermore, initial simulation issues such as incorrect capacitor value and oscilloscope settings affected early observations but were later corrected. Overall, hardware and simulated results followed the same trend and were consistent with theoretical expectations of clipper and clamper circuits.

7. Conclusion:

The experiment was performed using diode-based clipper and clamper circuits in simulation. The expected waveform modifications, including clipping and voltage shifting, were observed accurately. Minor differences between simulated and practical results were observed. Overall, the results were consistent with theoretical principles, confirming the proper functioning of the designed circuits.

8. Reference:

- [1] Department of Electrical and Electronic Engineering, EEE 2104: Electronic Devices Lab- Experiment 3, Faculty of Engineering, American International University- Bangladesh, pp.1-7.
- [2] Robert L. Boylestad, Louis Nashelsky, Electronic Devices and Circuit Theory, 9th Edition, 2007-2008
- [3] Adel S. Sedra, Kenneth C. Smith, Microelectronic Circuits, Saunders College Publishing, 3rd ed., ISBN: 0-03 051648-X, 1991.
- [4] David J. Comer, Donald T. Comer, Fundamentals of Electronic Circuit Design, John Wiley & Sons Canada, Ltd., ISBN: 0471410160, 2002.
- [5] Resistor values: <https://www.eleccircuit.com/how-to-basic-use-resistor/>, accessed on 20 September 2023.